

**HEMODYNAMIC**

NSAID PLUMBER

**TRIPLE WHAMMY**

BLOCKS BANKER

BILATERAL RAS

BILATERAL RAG

AFFERENT

EFFERENT

DEHYDRATION = WORSE

TRIPLE WHAMMY SABOTEUR

BLOCKS PG.

**TUBULAR ATN**

"PROXIMAL TUBULE"

PROXIMAL SEGMENT

WIDE - OPEN DRAINING OUT

OTHER NEPHROTOXINS

HIGH TROUGH

OTHER NEPHROTOXINS

CIRCULATING VOLUME

AMINOGLYCOSIDES

5-7 DAYS

5-7 Mg

NON-OLIGURIC

CORTX

VANCO

PROXIMAL TUBULE

AMPHO B

POLYURIA

LOW Mg

LOW Ca

NON-AG ACIDOSIS

**CRYSTAL**

CRYSTALS BLOCK FLOW - OBSTRUCTION AKI

ACYCLOVIR IV BOLUS

ALSO: CRYSTAL

GEMCITABINE

BEVACIZUMAB

HTN

ADAMTS13 NORMAL

TACROLIMUS

VASOCONSTRICTION

CYCLOSPORINE

**TMA**

ELECTROLYTE CHAOS

Ca Mg K Phos

ENDOTHELIAL TOXIN

INTERMITTENT C

GLOMERULAR MICROVASCULATURE

STRIPED FIBROSIS

**NEPHROTOXIN FACTORY**

AFTER STOPPED

DOSE-LIMITING

FLANK PAIN FIGURE

HYDRATION REDUCES RISK

PROXIMAL TUBULE

FANCONI SYNDROME

HYPOKALEMIA

HYPERKALEMIA

PROTEINURIA

LOW PHOS

AMINO ACIDS

TYPE II RTA

POLYURIA



## 1. Nephrotoxin Factory

### WHAT YOU SEE

Kidney gears, NEPHROTOXIN FACTORY banner

### WHAT IT ENCODES

Master scene organizing drug/toxin-induced AKI into five mechanisms: (1) hemodynamic (altered glomerular autoregulation), (2) tubular ATN (direct epithelial toxicity), (3) crystal nephropathy (intratubular obstruction), (4) thrombotic microangiopathy (endothelial injury), and (5) dose-limiting/preventable injury (contrast, pigment). Recognizing the mechanism predicts the timing, the urine sediment, and the prevention/treatment strategy on exam questions.

### BOARD TRAP

Don't lump all drug nephrotoxicity as 'ATN' — the discriminator is mechanism: hemodynamic AKI has a bland sediment and reverses with stopping the drug, ATN shows muddy-brown casts, crystal nephropathy shows crystalluria, and TMA shows schistocytes/thrombocytopenia.

## 2. Hemodynamic workshop

### WHAT YOU SEE

HEMODYNAMIC plumber, TRIPLE WHAMMY sign

### WHAT IT ENCODES

Hemodynamic AKI results from drugs that impair the kidney's ability to autoregulate glomerular pressure. The 'triple whammy' = NSAID + ACE-I/ARB + diuretic: the NSAID removes afferent vasodilation, the ACE-I/ARB removes efferent vasoconstriction, and the diuretic drops volume — together they collapse glomerular filtration pressure. This is functional (prerenal-type) AKI with a BLAND urine sediment and FENa <1%, and it reverses when the offending drug is stopped.

### BOARD TRAP

WRONG to call triple-whammy AKI 'ATN' — there is no tubular cell death, the sediment is bland, and creatinine rebounds within days of stopping the drugs. Treating it like ATN (just supportive waiting) misses the immediate fix: hold the NSAID/ACE-I and rehydrate.

## 3. NSAID afferent

### WHAT YOU SEE

NSAID plumber, afferent/efferent pipes, dehydration worse

### WHAT IT ENCODES

NSAIDs inhibit cyclooxygenase, blocking the vasodilatory prostaglandins (PGE2, PGI2) that dilate the AFFERENT arteriole. In volume-depleted states (CHF, cirrhosis, dehydration, true hypovolemia) renal perfusion is prostaglandin-dependent, so NSAIDs cause afferent constriction → fall in GFR. NSAIDs can also cause acute interstitial nephritis and minimal-change/membranous nephropathy (nephrotic syndrome).

### BOARD TRAP

The dangerous patient is the dehydrated/CHF/cirrhotic one — in a euvolemic healthy person NSAIDs rarely cause AKI, so the vignette discriminator is an NSAID started on top of volume depletion or diuretics.

## 4. ACE-I/ARB efferent

### WHAT YOU SEE

Triple whammy saboteur 'blocks PG', bilateral RAS

### WHAT IT ENCODES

ACE-inhibitors/ARBs block angiotensin-II-mediated constriction of the EFFERENT arteriole, lowering intraglomerular pressure (this is renoprotective long-term in CKD/diabetes). But when glomerular filtration depends on efferent constriction — bilateral renal artery stenosis (or stenosis of a solitary kidney) — these drugs cause an abrupt GFR drop. A creatinine rise up to ~30% after starting is acceptable and expected; more than 30% warrants stopping and evaluating for RAS.

### BOARD TRAP

WRONG to permanently discontinue an ACE-I for a small (<30%) creatinine bump — that rise reflects intended efferent dilation, not nephrotoxicity. But a sharp >30% rise, especially with hyperkalemia, should prompt suspicion of bilateral renal artery stenosis.

## 5. Tubular ATN workshop

### WHAT YOU SEE

TUBULAR ATN banker, 'proximal tubule wide-open draining'

### WHAT IT ENCODES

Acute tubular necrosis from direct nephrotoxins injures the proximal tubule (and thick ascending limb) epithelium. Damaged tubules can't reabsorb sodium, so FENa is >2% (>1% with urine Na >20) and urine osmolality is isosthenuric (~300). The pathognomonic sediment is muddy-brown (pigmented) granular casts and renal tubular epithelial cells. Recovery follows the ATN phases: initiation → maintenance (oliguric, the dangerous electrolyte phase) → recovery (polyuric).

### BOARD TRAP

FENa <1% does NOT exclude ATN if the patient is on diuretics (use FEurea <35% instead) or in early/nonoliguric contrast and pigment nephropathy — these ATN variants can paradoxically show a low FENa.

## 6. Aminoglycosides 5-7d

### WHAT YOU SEE

AMINOGLYCOSIDES, '5-7 days', NON-OLIGURIC

### WHAT IT ENCODES

Aminoglycosides (gentamicin, tobramycin, amikacin) accumulate in proximal tubule cells and cause ATN classically 5–7 days into therapy. It is typically NON-OLIGURIC and accompanied by potassium and magnesium wasting. Risk rises with prolonged courses, high troughs, and concurrent nephrotoxins; once-daily dosing and trough monitoring reduce risk because toxicity is trough-dependent while killing is peak-dependent.

### BOARD TRAP

WRONG answer is 'oliguric AKI on day 1' — aminoglycoside nephrotoxicity is delayed (5–7 days) and non-oliguric. Day-1 oliguria points elsewhere (contrast, ischemia, obstruction).



## 7. Vanco / Ampho B

### WHAT YOU SEE

VANCO and AMPHO B at proximal tubule bench, frog low Ca

### WHAT IT ENCODES

Vancomycin causes dose-dependent ATN, markedly potentiated when combined with aminoglycosides or piperacillin-tazobactam; keep troughs/AUC in range and monitor creatinine. Amphotericin B (especially the deoxycholate form) is intensely nephrotoxic via afferent vasoconstriction plus distal tubular membrane injury, causing potassium and magnesium wasting and a distal (type I) RTA; liposomal formulations and saline preloading reduce toxicity.

### BOARD TRAP

Don't forget amphotericin causes BOTH hypokalemia/hypomagnesemia AND a distal RTA — and vancomycin + piperacillin-tazobactam together raise AKI risk more than either alone, a common 'which combination is worst' exam point.

## 8. Crystal workshop

### WHAT YOU SEE

CRYSTAL station, 'crystals block flow - obstruction AKI'

### WHAT IT ENCODES

Crystal nephropathy occurs when drugs/metabolites precipitate inside tubules and cause intrarenal obstruction. Culprits: acyclovir (IV, in dehydration), sulfonamides, indinavir/atazanavir, methotrexate (high-dose), and uric acid/calcium phosphate in tumor lysis syndrome. Urine shows crystals; prevention is aggressive IV hydration to maintain high tubular flow, plus urinary alkalinization for methotrexate and uric acid.

### BOARD TRAP

Tumor lysis syndrome AKI: the discriminator labs are HIGH uric acid, HIGH potassium, HIGH phosphate, and LOW calcium — treat/prevent with hydration plus rasburicase (not just allopurinol, which only blocks new uric acid and doesn't clear existing load).

## 9. TMA workshop

### WHAT YOU SEE

TMA banner, 'electrolyte chaos Ca up K Phos', endothelial toxin

### WHAT IT ENCODES

Drug-induced thrombotic microangiopathy injures the glomerular/arteriolar endothelium → platelet microthrombi → microangiopathic hemolytic anemia (schistocytes), thrombocytopenia, and AKI. Key distinguishing point from TTP: ADAMTS13 activity is NORMAL in drug-induced TMA (it is severely deficient, <10%, in TTP). Management is stopping the offending drug; plasma exchange helps TTP, not most drug-induced TMA.

### BOARD TRAP

WRONG to assume ADAMTS13 is deficient in drug-induced TMA — it is normal. Severe ADAMTS13 deficiency (<10%) defines TTP and is the indication for urgent plasma exchange; checking ADAMTS13 is how you separate the two MAHA syndromes.

## 10. TMA drugs

### WHAT YOU SEE

GEMCITABINE, BEVACIZUMAB, glomerular fibrosis

### WHAT IT ENCODES

Drugs that classically cause TMA: gemcitabine (chemotherapy), VEGF inhibitors (bevacizumab — also cause hypertension and proteinuria via podocyte/endothelial injury), calcineurin inhibitors (tacrolimus, cyclosporine), quinine, and ticlopidine/clopidogrel. VEGF blockade is the mechanistic clue linking anti-angiogenic cancer therapy to hypertension, proteinuria, and TMA.

### BOARD TRAP

New hypertension + proteinuria + falling platelets in a cancer patient on bevacizumab is drug-induced TMA, not 'just chemo side effects' — recognizing it triggers holding the VEGF inhibitor.

## 11. Calcineurin inhibitors

### WHAT YOU SEE

TACROLIMUS / CYCLOSPORINE figures, vasoconstriction

### WHAT IT ENCODES

Calcineurin inhibitors (tacrolimus, cyclosporine) cause AKI by two mechanisms: acute, dose-dependent, reversible AFFERENT arteriolar vasoconstriction (functional/hemodynamic — improves with dose reduction) and chronic, irreversible TMA/interstitial fibrosis with arteriolar hyalinosis. They also cause hypertension, hyperkalemia (type IV RTA), and hypomagnesemia. Levels guide dosing because toxicity is concentration-dependent.

### BOARD TRAP

In a transplant patient with rising creatinine, the discriminator between CNI toxicity and acute rejection is the drug level (high → toxicity) and often biopsy — empirically increasing immunosuppression for presumed rejection when it's actually CNI toxicity worsens the kidney.

## 12. Dose-limiting workshop

### WHAT YOU SEE

DOSE-LIMITING flank-pain figure, 'hydration reduces risk'

### WHAT IT ENCODES

Preventable/dose-related injuries: contrast-associated nephropathy and pigment nephropathy (rhabdomyolysis-induced myoglobin, hemolysis-induced hemoglobin). Risk is reduced by IV isotonic saline hydration, minimizing contrast dose, and avoiding concurrent nephrotoxins. Contrast nephropathy peaks at 3–5 days and usually recovers; pigment nephropathy needs aggressive volume to flush the tubules.

### BOARD TRAP

Routine N-acetylcysteine and sodium bicarbonate for contrast prophylaxis are NOT proven superior to IV saline alone (PRESERVE trial) — the right answer for contrast nephropathy prevention is isotonic saline hydration, not NAC.



### 13. FENa over 1

#### WHAT YOU SEE

Low Ca / type II / polyuria flashcards, FENa clue

#### WHAT IT ENCODES

ATN urine indices: FENa >2% (FEurea >50% if on diuretics), urine Na >40, urine osmolality near serum (~300, isosthenuric), and muddy-brown granular casts. Proximal tubule injury can produce a Fanconi-type picture: glucosuria, phosphaturia, aminoaciduria, and a proximal (type II) RTA. Recovery often passes through a polyuric phase with ongoing electrolyte wasting.

#### BOARD TRAP

FENa <1% suggests PRERENAL azotemia, but several ATN variants (contrast, pigment, sepsis, and any ATN on diuretics) break this rule — confirm with the sediment (muddy-brown casts = ATN) rather than relying on FENa alone.

### 14. Acute interstitial nephritis

#### WHAT YOU SEE

Warning triangle / 'also crystal' near antibiotic vials in the crystal station

#### WHAT IT ENCODES

A sixth high-yield drug mechanism worth flagging: acute interstitial nephritis (AIN), an allergic tubulointerstitial reaction to drugs — classically beta-lactams, NSAIDs, sulfonamides, and PROTON PUMP INHIBITORS (an increasingly common cause). The classic triad of fever, rash, and eosinophilia is present in a minority; urine shows WBCs, WBC casts, and sometimes eosinophiluria. Treatment is stopping the drug, with corticosteroids if no rapid recovery.

#### BOARD TRAP

The full triad (fever + rash + eosinophilia) is present in fewer than 10–15% of AIN cases, so its ABSENCE does not exclude AIN — and don't forget PPIs/NSAIDs, the 'silent' causes without a dramatic allergic picture.